Aarnav Noble

416-474-7448 | anoble@uwaterloo.ca | LinkedIn | GitHub | Website

Education

University of Waterloo — Waterloo, Ontario

B.ASc. in Honours Computer Engineering — *Sep 2023 – Apr 2028 (expected)*

- Relevant Coursework: Operating Systems, Computer Architecture, Compilers, Databases, Computer Networks, Data Structures & Algorithms, Concurrency, Probability & Statistics, Linear Algebra

Experience

Member of Technical Staff Intern — Jan 2026 – Apr 2026

AethexAI — *San Francisco, California*

- Owned the customer-facing API and Python SDK end-to-end, covering **100+** endpoints including mid-call control
- Managed the production AWS/EKS cloud platform on Kubernetes, Terraform, and CI/CD, scaling to **500+** pods
- Kept streaming latency **sub-500ms** at **75+** concurrent calls on a platform handling **15K+** calls a day
- Built ML eval infrastructure with an LLM judge over **270** real calls, Langfuse tracing, and a gated TTS pipeline
- Designed a **5-phase** interrupt state machine across **3** transports, eliminating failures on ∼**80%** of interrupts
- Rebuilt agent RAG for launch with thread-safe ingestion, cross-turn dedup, and grounding over **10K+** chunks
- Automated regression testing with a Monte Carlo and WebRTC voice-agent simulator running **1,400** calls

AI/ML Developer — May 2025 – Aug 2025

Environment and Climate Change Canada — *Burlington, Ontario*

- Designed and trained a PyTorch GNN with **4** message-passing layers over a **10K+** node graph
- Beat gradient-boosted baselines by **85%**, replacing hand-built spatial features with learned graph structure
- Extended usable output **5x** by generating forecasts at graph nodes with no observed data
- Engineered **50+** spatial-temporal features and cut a full run from hours to minutes by vectorizing and parallelizing
- Ran region-holdout cross-validation and a **100+** run hyperparameter sweep to kill spatial overfitting
- Led ML on a **6-person** research team from architecture to eval, presenting results toward a publication

Software Engineering Intern — Sep 2024 – Dec 2024

Toronto Transit Commission (TTC) — *Toronto, Ontario*

- Built a config-as-code deployment pipeline enforcing one software state across a **10K+** device fleet
- Automated zero-touch bare-metal provisioning with network-boot and Bash, cutting setup roughly **50%**
- Wrote Bash orchestration for staged patch and driver rollouts across **2K+** machines, with auto-retry on failure

Software Developer — Jan 2024 – Apr 2024

Ontario Chamber of Commerce — *Toronto, Ontario*

- Built **10+** reusable React components with schema-driven config panels, editable by non-devs in a visual editor
- Fed external data into **15+** Shopify Metaobject types from a Go service on Kubernetes, bound to live content
- Secured the app with session-token auth, scoped OAuth, and HMAC-verified webhooks against forged events

Projects

vLLM AIBrix – Top-20 Contributor | *Go, Kubernetes, Redis, Prometheus* |

- Built early PD admission that reserves decode capacity before prefill, cutting burst GPU prefills from **16** to **1**
- Redesigned KPA burst scaling around persistent panic state, extending peak-capacity holds from **0s** to **180s**
- Added per-pod expiry to the prefix-cache index, so routing skips pods that evicted their KV cache

Vestige | *Python, Go, TypeScript, Next.js, OpenTelemetry, ClickHouse* |

- Built a Go service ingesting agent LLM and tool-call spans at **30K+**/sec, with zero loss in a ClickHouse outage
- Shipped a Python SDK and replay engine that reruns **50-span** agent failures from fixtures in under **5s**
- Gated PRs on replayed regression evals in a no-network sandbox, plus a **100-pair** tenant-isolation test

Skills

Languages: Python, C/C++, Go, TypeScript, JavaScript, Java, SQL, Bash
Systems & Backend: Kubernetes, Docker, Terraform, AWS, GCP, Git, CI/CD, Linux, Networking, FastAPI, React, Next.js, Node.js, REST APIs, GraphQL, PostgreSQL, Redis, Caching, ClickHouse, OpenTelemetry, Prometheus
ML & Data: PyTorch, TensorFlow, NumPy, pandas, Scikit-learn, XGBoost
Concepts: Distributed Systems, System Design, Real-time Systems, Scalability, Reliability, Machine Learning, LLM Inference, AI Agents, Agent Evals, RAG, Fine-tuning, Observability, Testing, Security, API Design